

The Geometry of the Virtue of Complexity in Asset Pricing

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Abstract

Recent debates in empirical asset pricing question whether the performance gains of high-dimensional machine learning models reflect genuine economic structure or statistical illusion. We argue that this debate can be reframed geometrically. Regardless of nonlinear exposure maps or high-dimensional feature expansions, most asset pricing models ultimately produce low-dimensional return spans. We show that the economically relevant object is a point on a Grassmann manifold, and that the “virtue of complexity” can be evaluated through subspace stability, intrinsic dimension, and curvature diagnostics. This geometric perspective separates parameter complexity from structural complexity and clarifies when additional flexibility reflects genuine discovery rather than statistical illusion.

1 Introduction

Recent work in empirical asset pricing argues that nonlinear and high-dimensional machine learning models can substantially improve the estimation of stochastic discount factors (SDFs) and conditional factor structures. Critics counter that such gains may reflect overfitting or statistical illusion rather than genuine economic structure.

We propose to reframe this debate geometrically.

Most conditional factor models, regardless of how exposures are parameterized, ultimately produce a low-rank representation of returns. Let N denote the number of assets and k the number of latent factors. At each time t , cross-sectional returns $r_t \in \mathbb{R}^N$ are modeled as

$$r_t \approx \beta_t^\top f_t,$$

where $\beta_t \in \text{Mat}_{k,N}(\mathbb{R})$ and $f_t \in \mathbb{R}^k$.

The economically relevant object is therefore not the parameter vector used to generate exposures, but the k -dimensional return span

$$V_t = \text{span}(\beta_t^\top) \in \text{Gr}(k, N),$$

a point on the Grassmann manifold of k -dimensional subspaces of $\mathbb{R}^N$. As time evolves, the model induces a trajectory

$$t \longmapsto V_t$$

in $\text{Gr}(k, N)$.

This object is invariant under rotations of the factor basis and under many reparameterizations of the exposure map. Distinct parameter vectors may therefore generate identical return subspaces. From an economic perspective, the trajectory of subspaces is fundamental; the specific parameterization used to construct it is secondary.

The central question becomes:

Does increasing functional complexity enlarge the identified trajectory on $\text{Gr}(k, N)$ in a stable and persistent manner?

In this formulation, model flexibility enlarges the admissible family of subspace trajectories, while the rank parameter k fixes their intrinsic dimension. Complexity is *virtuous* if it expands the identifiable return span in a stable way. It is *illusory* if it merely introduces weakly identified directions that rotate under sampling variation without increasing intrinsic dimensionality.

This geometric viewpoint separates *parameter complexity* from *structural complexity*. The former concerns the size and nonlinearity of the parameter space; the latter concerns the dimension, curvature, and stability of the induced return subspaces on $\text{Gr}(k, N)$.

Our contribution is to formalize this distinction. We show that projection-based estimators in conditional factor models depend only on induced Grassmann trajectories, and we characterize identification through curvature of the projection functional. In a stylized population setting, we prove that when the nominal dimension k exceeds the true signal dimension, the objective exhibits flat directions corresponding to rotations within a noise eigenspace. These directions manifest empirically as subspace instability and collapse of effective rank.

This framework provides structural diagnostics for the virtue of complexity: genuine increases in intrinsic dimensionality correspond to persistent expansions of the return subspace, while illusory gains correspond to geometrically flat or nearly flat directions that are unstable under perturbations. Rather than evaluating complexity through parameter counts or short-horizon predictive gains alone, we propose to assess it through Grassmann invariants, such as intrinsic dimension, subspace stability, and curvature.

2 IPCA as a Prototype: From Parameters to Subspaces

We use the linear IPCA model [6] as a transparent prototype for illustrating the geometric structure underlying conditional factor models. We follow here the arguments developed in [8].

2.1 Objective function of IPCA

At each time t , cross-sectional returns $r_t \in \mathbb{R}^N$ are modeled as

$$r_t = \beta_t^\top f_t + \varepsilon_t, \quad (1)$$

where $f_t \in \mathbb{R}^k$ are latent factors and exposures are driven by characteristics $Z_t \in \text{Mat}_{N,m}(\mathbb{R})$ through

$$\beta_t = \Gamma Z_t^\top, \quad \Gamma \in \text{Mat}_{k,m}(\mathbb{R}). \quad (2)$$

Equivalently,

$$r_t = Z_t \Gamma^\top f_t + \varepsilon_t. \quad (3)$$

The matrix Γ parameterizes the exposure map from characteristics to factor loadings.

The model is invariant under orthogonal rotations in factor space: for any $O \in O(k)$,

$$\begin{aligned} \Gamma &\mapsto O\Gamma, \\ f_t &\mapsto Of_t, \end{aligned}$$

leaves $Z_t \Gamma^\top f_t$ unchanged. Hence Γ is identified only through its row space

$$\text{row}(\Gamma) = \text{span}(\Gamma^\top) \in \text{Gr}(k, m).$$

Thus the parameter space carries an intrinsic Grassmann geometry.

The standard IPCA estimator solves

$$\min_{\Gamma, \{f_t\}} \frac{1}{T} \sum_{t=1}^T \|r_t - Z_t \Gamma^\top f_t\|^2.$$

Profiling out the factors f_t yields

$$f_t^*(\Gamma) = (\Lambda_t^\top \Lambda_t)^{-1} \Lambda_t^\top r_t, \quad (4)$$

where

$$\Lambda_t(\Gamma) = Z_t \Gamma^\top, \quad (5)$$

and the concentrated objective function becomes

$$\mathcal{L}(\Gamma) = \frac{1}{T} \sum_{t=1}^T \|(I - P_t(\Gamma))r_t\|^2, \quad (6)$$

where

$$P_t(\Gamma) = \Lambda_t(\Lambda_t^\top \Lambda_t)^{-1} \Lambda_t^\top$$

is the orthogonal projector onto

$$\begin{aligned} V_t(\Gamma) &= \text{span}(Z_t \Gamma^\top) \\ &= \text{span}(\beta_t^\top) \in \text{Gr}(k, N). \end{aligned}$$

2.2 IPCA objective function as a Grassmann functional

The crucial observation is that $\mathcal{L}(\Gamma)$ depends on Γ only through the induced return subspaces $V_t(\Gamma)$. In particular, the identified parameter of IPCA lies in $\text{Gr}(k, m)$ rather than in $\text{Mat}_{k,m}(\mathbb{R})$. Therefore IPCA may be viewed as minimizing a projection functional

$$\mathcal{J}(V) = \frac{1}{T} \sum_{t=1}^T \|(I - \Pi_{V_t})r_t\|^2 \quad (7)$$

over the admissible family

$$\mathcal{A} = \{(V_t) : V_t = \text{span}(Z_t \Gamma^\top) \text{ for some } \Gamma\}.$$

Thus the economically relevant object is not the parameter matrix Γ , but the trajectory of return subspaces

$$t \longmapsto V_t(\Gamma) \in \text{Gr}(k, N).$$

This reformulation makes explicit that IPCA already operates on the Grassmann manifold: parameter complexity affects the set of attainable subspaces, while the objective itself measures only projection onto k -dimensional return spans.

3 Complexity in Conditional Factor Models

The IPCA specification in Section 2 is a special case of a broader class of *conditional factor* models in which exposures are generated from characteristics through a possibly nonlinear map. We write the general specification as

$$\begin{aligned} r_t &\approx \beta_t^\top f_t, \\ \beta_t &= F_\theta(Z_t), \end{aligned} \tag{8}$$

where $Z_t \in \text{Mat}_{N,m}(\mathbb{R})$ collects characteristics, $f_t \in \mathbb{R}^k$ are latent factors, and θ denotes model parameters.

3.1 Return subspaces and structural complexity

Regardless of the parameterization F_θ , the fitted returns lie in a k -dimensional subspace

$$V_t(\theta) = \text{span}(\beta_t^\top) \in \text{Gr}(k, N). \tag{9}$$

Thus model complexity enlarges the parameter space for θ , but the economically relevant object remains the trajectory

$$t \longmapsto V_t(\theta) \text{ in } \text{Gr}(k, N).$$

This separates:

- (A) *Parameter complexity*: dimension and nonlinearity of θ ,
- (B) *Structural complexity*: the set of attainable Grassmann trajectories and their stability.

In this formulation, the “virtue of complexity” means that increasing flexibility enlarges the *identified* return subspace in a stable manner, rather than introducing weak directions that rotate under sampling noise.

3.2 Rotation invariance

As in IPCA, the representation is invariant under orthogonal factor rotations:

$$\begin{aligned} \beta_t &\mapsto O\beta_t, \\ f_t &\mapsto Of_t, \end{aligned}$$

for $O \in O(k)$. Hence any such model is identified only through the row space of β_t , equivalently through $V_t(\theta) \in \text{Gr}(k, N)$. This invariance explains why subspace based diagnostics are intrinsic and basis free.

3.3 Lifted linear models and Random Fourier Features

An important intermediate class between linear IPCA and fully nonlinear neural networks is obtained by *feature lifting*:

$$\beta_t = \Gamma \Phi(Z_t)^\top, \quad \Gamma \in \text{Mat}_{k, \tilde{m}}(\mathbb{R}), \tag{10}$$

where $\Phi(Z_t) \in \text{Mat}_{N, \tilde{m}}(\mathbb{R})$ is a high-dimensional feature matrix. When $\tilde{m} \gg m$, this yields a high-parameter model while preserving the same rank k return representation

$$V_t(\Gamma) = \text{span}(\Phi(Z_t)\Gamma^\top) \in \text{Gr}(k, N).$$

A canonical choice for Φ is provided by *Random Fourier Features* [9], [10]. For characteristics $z_{i,t} \in \mathbb{R}^m$, define

$$\phi(z) = \sqrt{\frac{2}{\tilde{m}}} \left(\cos(\omega_1^\top z + b_1), \dots, \cos(\omega_{\tilde{m}}^\top z + b_{\tilde{m}}) \right)^\top,$$

with $\omega_j \sim p(\omega)$ and $b_j \sim \text{Unif}[0, 2\pi]$. Stacking $\phi(z_{i,t})$ yields

$$\Phi(Z_t) = \begin{pmatrix} \phi(z_{1,t})^\top \\ \vdots \\ \phi(z_{N,t})^\top \end{pmatrix}.$$

The resulting RFF conditional factor model is

$$r_t \approx \Phi(Z_t) \Gamma^\top f_t. \quad (11)$$

3.4 Profiled objective function for lifted linear models

As in IPCA, factors can be profiled out explicitly. Let

$$\Lambda_t(\Gamma) = \Phi(Z_t) \Gamma^\top \in \text{Mat}_{N,k}(\mathbb{R}).$$

Then

$$f_t^*(\Gamma) = (\Lambda_t^\top \Lambda_t)^{-1} \Lambda_t^\top r_t,$$

and the concentrated objective becomes

$$\mathcal{L}(\Gamma) = \frac{1}{T} \sum_{t=1}^T \|(I - P_t(\Gamma)) r_t\|^2, \quad (12)$$

where

$$P_t(\Gamma) = \Lambda_t(\Gamma) (\Lambda_t^\top \Lambda_t)^{-1} \Lambda_t(\Gamma)^\top$$

is the orthogonal projector onto

$$V_t(\Gamma) = \text{span}(\Lambda_t(\Gamma)).$$

Thus even after nonlinear lifting, estimation reduces to minimizing a projection functional over k -dimensional subspaces in $\mathbb{R}^N$. The only effect of lifting is therefore to enlarge the admissible family of subspaces $\{V_t\}$, not to change the geometric nature of the objective.

3.5 Why lifting changes flexibility but not intrinsic dimension

Feature lifting enlarges the admissible family

$$t \mapsto V_t(\Gamma) \in \text{Gr}(k, N)$$

through a richer design matrix $\Phi(Z_t)$. However, the objective (12) still measures projection onto k -dimensional spans. Hence additional flexibility is structurally meaningful only if it yields:

- (i) persistent increases in intrinsic dimension (e.g. higher effective rank),
- (ii) stable Grassmann trajectories across samples.

Otherwise, additional directions correspond to weakly identified components, leading to subspace instability and effective-rank collapse.

3.6 Fully nonlinear exposure maps

A fully nonlinear alternative specifies

$$\beta_t = F_\theta(Z_t),$$

with F_θ implemented by a neural network. Examples include multilayer perceptrons, recurrent networks (e.g. LSTMs), and other deep architectures, but the geometric conclusions below are independent of the specific implementation of F_θ . Although parameter-space geometry becomes highly nonlinear, the induced return spans remain

$$V_t(\theta) = \text{span}(\beta_t^\top) \in \text{Gr}(k, N).$$

Thus, across IPCA, lifted linear models, and neural networks, model complexity enlarges the admissible family of Grassmann trajectories but leaves the intrinsic rank parameter k unchanged. **The virtue of complexity must therefore be evaluated through Grassmann invariants of $\{V_t\}$, such as intrinsic dimension, stability, and curvature, rather than through raw parameter counts.**

4 Return Subspaces and the Grassmann Manifold

4.1 Intrinsic Dimensionality

We recall that the *effective (entropic) rank* $\text{erank}(\Sigma)$ of a positive definite matrix $\Sigma \in \text{Mat}_n(\mathbb{R})$ is defined as follows [11]. Let $\lambda_1 \geq \dots \lambda_n \geq 0$ denote the eigenvalues, and let

$$p_i = \frac{\lambda_i}{\text{tr}(\Sigma)},$$

for $i = 1, \dots, n$. Then

$$\text{erank}(\Sigma) = \exp(H(p)),$$

where $H(p) = -\sum_{i=1}^n p_i \log p_i$ is the Shannon entropy of the discrete probability distribution p . Note that, in particular, (i) if all eigenvalues are equal, $p_i = 1/n$, then $H(p) = \log n$, and so $\text{erank}(\Sigma) = n$, (ii) if one eigenvalue dominates, $p_1 \approx 1$, then $H(p) \approx 0$, so that $\text{erank}(\Sigma) \approx 1$.

Let $\hat{f}_t$ be fitted factors and define the *estimated factor covariance matrix*:

$$\hat{\Sigma}_f = \frac{1}{T} \sum_{t=1}^T \hat{f}_t \hat{f}_t^\top. \quad (13)$$

Definition 4.1. The intrinsic dimension of a fitted return representation is measured by the effective rank $\text{erank}(\hat{\Sigma}_f)$ of the estimated factor covariance matrix.

If $\text{erank}(\hat{\Sigma}_f) \ll k$, then only a few factors are effectively active (the remainder are weakly identified or noise-fitting). Tracking $\text{erank}(\hat{\Sigma}_f)$ across rolling windows provides a stability diagnostic: if additional nonlinear flexibility primarily fits noise, the spectrum of $\hat{\Sigma}_f$ will exhibit rapid decay and the effective rank will remain small.

4.2 Subspace Stability

Let V_1 and V_2 denote two k -dimensional subspaces of $\mathbb{R}^N$. Choose orthonormal basis matrices $U_1, U_2 \in \text{Mat}_{N,k}(\mathbb{R})$ such that

$$\begin{aligned} V_i &= \text{span}(U_i), \\ U_i^\top U_i &= I_k, \end{aligned}$$

for $i = 1, 2$. We shall define several standard concepts of the distance between the subspaces V_1 and V_2 [?]. All these distance functions can be expressed explicitly in terms of the *principal angles*.

The principal angles $\theta_1, \dots, \theta_k \in [0, \pi/2]$ between V_1 and V_2 are defined by

$$\cos \theta_i = \sigma_i(U_1^\top U_2),$$

$i = 1, \dots, k$, where $\sigma_1 \geq \dots \geq \sigma_k \geq 0$ are the singular values. Equivalently, if

$$U_1^\top U_2 = Q_1 \operatorname{diag}(\sigma_1, \dots, \sigma_k) Q_2^\top \quad (14)$$

is the singular value decomposition of $U_1^\top U_2$, then $\theta_i = \arccos(\sigma_i)$.

Geodesic distance. Equip $\operatorname{Gr}(k, N)$ with the canonical Riemannian metric induced from the quotient $\operatorname{St}(k, N)/O(k)$ (equivalently, the metric induced by the Frobenius inner product on horizontal tangent vectors). The geodesic Grassmann distance between V_1 and V_2 is defined intrinsically as

$$d_{\text{geo}}(V_1, V_2) = \inf_{\gamma} \operatorname{Length}(\gamma), \quad (15)$$

where the infimum is taken over all piecewise smooth curves $\gamma : [0, 1] \rightarrow \operatorname{Gr}(k, N)$ such that $\gamma(0) = V_1$ and $\gamma(1) = V_2$, and $\operatorname{Length}(\gamma)$ is computed with respect to the canonical metric. Equivalently, $d_{\text{geo}}(V_1, V_2)$ is the length of the unique shortest geodesic connecting V_1 to V_2 .

In terms of the principal angles, the geodesic Grassmann distance admits the explicit form

$$\begin{aligned} d_{\text{geo}}(V_1, V_2) &= \left(\sum_{i=1}^k \theta_i^2 \right)^{1/2} \\ &= \left(\sum_{i=1}^k \arccos^2(\sigma_i(U_1^\top U_2)) \right)^{1/2}. \end{aligned} \quad (16)$$

In empirical work, alternative distances are often used.

Projection (Frobenius) distance. Let $P_i = U_i U_i^\top$ denote the orthogonal projector onto V_i . The projection distance is

$$d_{\text{proj}}(V_1, V_2) = \|P_1 - P_2\|_F. \quad (17)$$

In terms of principal angles,

$$\|P_1 - P_2\|_F^2 = 2 \sum_{i=1}^k \sin^2 \theta_i. \quad (18)$$

Chordal distance. The (scaled) chordal distance is defined by

$$\begin{aligned} d_{\text{chord}}(V_1, V_2) &= \|\sin \Theta\|_F \\ &= \left(\sum_{i=1}^k \sin^2 \theta_i \right)^{1/2}, \end{aligned} \quad (19)$$

where $\Theta = \operatorname{diag}(\theta_1, \dots, \theta_k)$. Combining (18) and (19),

$$d_{\text{proj}} = \sqrt{2} d_{\text{chord}}.$$

For small principal angles (i.e., when subspaces are close),

$$\sin \theta_i \approx \theta_i,$$

and therefore

$$d_{\text{chord}} \approx d_{\text{geo}}.$$

Hence all three distances are locally equivalent metrics on $\text{Gr}(k, N)$.

In empirical applications, d_{proj} and d_{chord} are often computationally convenient, while d_{geo} has the advantage of being intrinsically defined via the canonical Riemannian geometry of the Grassmannian.

4.3 Implications for Complexity

Increasing functional flexibility (for example, replacing Z_t by $\Phi(Z_t)$ with nonlinear features) enlarges the admissible family of subspaces that can be generated.

However, the objective (12) continues to measure only the projection of returns onto k -dimensional subspaces. Thus:

- Functional complexity increases the size of the admissible set in $\text{Gr}(k, N)$,
- But does not increase intrinsic dimension unless the minimizing subspaces expand in a stable manner.

This observation makes precise the distinction between functional complexity and geometric complexity and provides a direct geometric interpretation of the “virtue of complexity” debate.

5 A Stylized Population Theorem

We now formalize the geometric content of the “virtue of complexity” debate under a stylized population factor model. The main message is simple: in population, the projection error objective identifies the signal subspace. If the nominal dimension k exceeds the true signal dimension k_0 , the additional directions are *not identified* (flat directions on the Grassmannian).

5.1 Population setup

We consider the following stylized population factor model. Fix t and write the (conditional) factor structure

$$r_t = (\beta_t^0)^\top f_t + \varepsilon_t, \quad (20)$$

where $r_t \in \mathbb{R}^N$, the (true) loading matrix $\beta_t^0 \in \text{Mat}_{k_0, N}(\mathbb{R})$ has full row rank, $f_t \in \mathbb{R}^{k_0}$ are factors, and $\varepsilon_t \in \mathbb{R}^N$ are idiosyncratic shocks. Furthermore, we assume:

- (A1) $\mathbb{E}(\varepsilon_t) = 0$ and $\mathbb{E}(\varepsilon_t \varepsilon_t^\top) = \sigma^2 I_N$ for some $\sigma^2 > 0$;
- (A2) $\mathbb{E}(f_t) = 0$ and $\Sigma_f = \mathbb{E}(f_t f_t^\top) \in \text{Mat}_{k_0}(\mathbb{R})$ is positive definite;
- (A3) ε_t is uncorrelated with f_t (hence $\mathbb{E}(\varepsilon_t f_t^\top) = 0$).

For a candidate k -dimensional subspace $V \in \text{Gr}(k, N)$, let Π_V denote orthogonal projection onto V , and consider the population projection-error functional

$$\mathcal{J}(V) = \mathbb{E}(\|(I - \Pi_V)r_t\|^2), \quad (21)$$

where $\mathbb{E}(\cdot)$ denotes expected value.

5.2 Identification of the signal subspace

The theorem below shows that population minimizers are signal subspaces.

Theorem 5.1. *Let $S = \text{span}((\beta_t^0)^\top) \in \text{Gr}(k_0, N)$ be the signal subspace. Then for any $V \in \text{Gr}(k, N)$,*

$$\mathcal{J}(V) = \underbrace{\mathbb{E}(\|(I - \Pi_V)(\beta_t^0)^\top f_t\|^2)}_{\text{signal loss}} + \underbrace{\sigma^2(N - k)}_{\text{noise floor}}. \quad (22)$$

Moreover:

- (i) *If $k = k_0$, then $\mathcal{J}(V)$ is uniquely minimized at $V = S$.*
- (ii) *If $k \geq k_0$, then $\mathcal{J}(V)$ is minimized if and only if $S \subseteq V$. In particular, all minimizers have the same value*

$$\inf_{V \in \text{Gr}(k, N)} \mathcal{J}(V) = \sigma^2(N - k),$$

and the additional $(k - k_0)$ directions inside V are not identified in population.

Proof. Using (20) and expanding (21),

$$\begin{aligned} \mathcal{J}(V) &= \mathbb{E}(\|(I - \Pi_V)((\beta_t^0)^\top f_t + \varepsilon_t)\|^2) \\ &= \mathbb{E}(\|(I - \Pi_V)(\beta_t^0)^\top f_t\|^2) + \mathbb{E}(\|(I - \Pi_V)\varepsilon_t\|^2), \end{aligned}$$

where the cross term vanishes by (A3). Under (A1),

$$\begin{aligned} \mathbb{E}(\|(I - \Pi_V)\varepsilon_t\|^2) &= \mathbb{E}(\varepsilon_t^\top (I - \Pi_V) \varepsilon_t) \\ &= \text{tr}((I - \Pi_V) \mathbb{E}[\varepsilon_t \varepsilon_t^\top]) \\ &= \sigma^2 \text{tr}(I - \Pi_V) \\ &= \sigma^2(N - k), \end{aligned}$$

which yields (22). The signal term is zero if and only if $(I - \Pi_V)(\beta_t^0)^\top = 0$, i.e. $\text{span}((\beta_t^0)^\top) \subseteq V$. When $k = k_0$, this implies $V = S$. $\square$

The geometric interpretation of the flat directions is as follows. When $k > k_0$, Theorem 5.1 implies that the population objective is constant along the set

$$\{V \in \text{Gr}(k, N) : S \subseteq V\},$$

a positive-dimensional submanifold of $\text{Gr}(k, N)$. These directions correspond to rotations within the noise eigenspace and therefore incur no change in population fit. They are geometrically flat directions of the objective and become weakly identified under sampling variation.

The preceding theorem shows that the population IPCA objective identifies the signal subspace but exhibits flat directions when the nominal dimension exceeds the true signal dimension. This geometric degeneracy motivates the subspace-stability and effective-rank diagnostics introduced below.

5.3 Implication for “virtue of complexity”

Theorem 5.1 clarifies the key distinction: in population, improving functional flexibility can only help insofar as it enables the model to recover the *signal subspace* S more accurately. Increasing functional complexity does not, by itself, create new intrinsic dimensions unless the data generating process contains them.

In particular, if k is chosen larger than the true signal dimension k_0 , then population fit does not penalize the extra directions; their presence will typically manifest empirically as *subspace instability* and *low effective rank* of fitted factor covariance matrices.

6 Complexity and Curvature

High-dimensional nonlinear models often introduce weakly identified directions. In the Grassmann formulation, this manifests as:

- (i) Existence of near-flat directions of the objective function along Grassmann geodesics;
- (ii) Ill-conditioning of the Hessian of the objective function;
- (iii) Sensitivity to perturbations in the data.

Such geometric degeneracy is a signature of over-parameterization without structural gain.

6.1 Curvature degeneracy and effective-rank collapse

We now give a stylized theorem linking geometric degeneracy (vanishing curvature of the Grassmann objective in certain directions) to an effective-rank collapse of fitted factor covariances. The statement is deliberately idealized, but it captures the mechanism that makes “illusory” complexity unstable.

Fix t and write the population second moment of returns as

$$\Sigma_r = \mathbb{E}(r_t r_t^\top). \quad (23)$$

For $V \in \text{Gr}(k, N)$ with projector Π_V , consider the population Grassmann functional (equivalent to minimizing projection error)

$$\begin{aligned} \mathcal{J}(V) &= \mathbb{E}(\|(I - \Pi_V)r_t\|^2) \\ &= \text{tr}(\Sigma_r) - \text{tr}(\Pi_V \Sigma_r). \end{aligned} \quad (24)$$

Thus minimizing $\mathcal{J}$ is equivalent to maximizing $\text{tr}(\Pi_V \Sigma_r)$ over $V \in \text{Gr}(k, N)$.

The theorem below formalizes the link between curvature degeneracy and effective rank collapse. Assume the stylized population factor model as defined in Section 5.

Theorem 6.1. *Let $\lambda_1 \geq \dots \geq \lambda_{k_0} > \sigma^2$ denote the k_0 “signal” eigenvalues of $\Sigma_r = \beta^\top \Sigma_f \beta + \sigma^2 I_N$ above the noise floor, and $\lambda_{k_0+1} = \dots = \lambda_N = \sigma^2$ the noise eigenvalues. Let $V^* \in \text{Gr}(k, N)$ be any minimizer of $\mathcal{J}$ with $k \geq k_0$, so that $S = \text{span}(\beta^\top) \subseteq V^*$ (cf. Theorem 5.1). Then:*

- (i) Flat directions. *If $k > k_0$, there exist nonzero tangent directions $\xi \in T_{V^*}\text{Gr}(k, N)$ such that the Riemannian Hessian of $\mathcal{J}$ vanishes:*

$$\text{Hess}_{V^*} \mathcal{J}[\xi, \xi] = 0.$$

More precisely, the null space of the Hessian contains all infinitesimal rotations of the $(k - k_0)$ “extra” dimensions of V^ within the noise eigenspace of Σ_r .*

- (ii) Curvature is controlled by eigenvalue gaps. *For tangent directions that rotate a signal direction (eigenvalue λ_i) toward the noise eigenspace (eigenvalue σ^2), the Hessian quadratic form satisfies*

$$\text{Hess}_{V^*} \mathcal{J}[\xi, \xi] \asymp \sum_{i=1}^{k_0} (\lambda_i - \sigma^2) \|\xi_i\|^2, \quad (25)$$

for a suitable decomposition of ξ into principal rotation components ξ_i , where $a \asymp b$ means there exist constants $c_1, c_2 > 0$ such that $c_1 b \leq a \leq c_2 b$. Hence curvature degenerates as the gaps $\lambda_i - \sigma^2$ shrink.

(iii) Effective rank collapse for over-specified k . Consider the “population fitted factors” corresponding to a representative orthonormal basis $U \in \text{St}(k, N)$ of V^* :

$$\begin{aligned} f_t^* &= U^\top r_t \in \mathbb{R}^k, \\ \Sigma_f^* &= \mathbb{E}(f_t^* (f_t^*)^\top) \\ &= U^\top \Sigma_r U. \end{aligned} \tag{26}$$

Then Σ_f^* has the block spectral structure

$$\text{spec}(\Sigma_f^*) = \{\lambda_1, \dots, \lambda_{k_0}, \underbrace{\sigma^2, \dots, \sigma^2}_{k-k_0 \text{ times}}\},$$

so that the effective rank satisfies

$$\text{erank}(\Sigma_f^*) \leq k_0 + \Delta(\lambda, \sigma; k), \tag{27}$$

where the “excess” term $\Delta(\lambda, \sigma; k)$ is small whenever the additional eigenvalues are near the noise floor (e.g. when $\sigma^2 \ll \sum_{i=1}^{k_0} \lambda_i$ and $k - k_0$ is moderate). In particular, over-specifying k introduces $k - k_0$ weak components whose contribution to erank is negligible, i.e. effective rank collapses toward k_0 .

Proof. As a consequence of (24), minimizing $\mathcal{J}$ over $\text{Gr}(k, N)$ is equivalent to maximizing $F(V) = \text{tr}(\Pi_V \Sigma_r)$.

Step 1. Let $\mu_1 \geq \dots \geq \mu_N$ be the eigenvalues of Σ_r . By the Ky Fan maximum principle [3],

$$\sup_{V \in \text{Gr}(k, N)} F(V) = \sum_{i=1}^k \mu_i,$$

and the maximizers are precisely the k -dimensional subspaces spanned by eigenvectors associated with the top k eigenvalues. Since σ^2 has multiplicity $N - k_0$, the maximizer is non-unique whenever $k > k_0$, and the set of maximizers forms a Grassmann submanifold. In our model, $\mu_1, \dots, \mu_{k_0} = \lambda_1, \dots, \lambda_{k_0}$ and $\mu_{k_0+1} = \dots = \mu_N = \sigma^2$, hence for $k \geq k_0$ any maximizer V^* must contain the signal subspace $S = \text{span}(\beta^\top)$.

Step 2. Fix a maximizer V^* and choose an orthonormal basis $U \in \text{St}(k, N)$ for V^* , and an orthonormal complement $U_\perp \in \text{St}(N - k, N)$ so that $[U \ U_\perp]$ is orthogonal. Any tangent direction at V^* can be represented (non-uniquely) by a matrix $K \in \text{Mat}_{(N-k) \times k}(\mathbb{R})$ via the geodesic

$$U(\varepsilon) = (U + U_\perp K \varepsilon) (I_k + K^\top K \varepsilon^2)^{-1/2}$$

and so

$$\Pi(\varepsilon) = U(\varepsilon) U(\varepsilon)^\top,$$

where $\Pi(0) = \Pi_{V^*}$ and $\dot{\Pi}(0)$ corresponds to the tangent vector.

Write the block decomposition of Σ_r in this basis:

$$\Sigma_r = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

with

$$A_{11} = U^\top \Sigma_r U, \quad A_{22} = U_\perp^\top \Sigma_r U_\perp, \quad A_{12} = U^\top \Sigma_r U_\perp.$$

Since V^* is spanned by eigenvectors of Σ_r , we have $A_{12} = 0$. A standard second-order calculation gives

$$F(V(\varepsilon)) = F(V^*) - \varepsilon^2 \text{tr}(K^\top (A_{11} - A_{22})K) + o(\varepsilon^2). \quad (28)$$

(Thus the Riemannian Hessian of $-F$ at V^* is the quadratic form $K \mapsto \text{tr}(K^\top (A_{11} - A_{22})K)$.)

Step 3. When $k > k_0$, the subspace V^* contains $k - k_0$ directions inside the noise eigenspace (corresponding to the eigenvalue σ^2). Along tangent directions K that only rotate those noise directions within the noise eigenspace, both corresponding diagonal blocks of A_{11} and A_{22} equal $\sigma^2 I$, hence $A_{11} - A_{22}$ vanishes on those components and (28) yields zero curvature. Therefore there exist nonzero ξ with $\text{Hess}_{V^*} \mathcal{J}[\xi, \xi] = 0$. This proves part (i) of the theorem.

Step 4. Consider a tangent direction that mixes a signal eigenvector (corresponding to eigenvalue λ_i) inside V^* with a noise eigenvector (corresponding to eigenvalue σ^2) in $V^{*\perp}$. In the eigenbasis, A_{11} is diagonal with entries from $\{\lambda_1, \dots, \lambda_{k_0}, \sigma^2, \dots, \sigma^2\}$ and A_{22} has σ^2 on the noise components. Restricting (28) to these principal mixing components gives

$$\begin{aligned} \text{Hess}_{V^*} \mathcal{J}[\xi, \xi] &= 2 \text{tr}(K^\top (A_{11} - A_{22})K) \\ &\asymp \sum_{i=1}^{k_0} (\lambda_i - \sigma^2) \|\xi_i\|^2, \end{aligned}$$

which is (25). (The proportionality constants depend only on the chosen identification between K and ξ .) This proves part (ii).

Step 5. Let U be any orthonormal basis of V^* and define $f_t^* = U^\top r_t$. Then $\Sigma_f^* = \mathbb{E}(f_t^* (f_t^*)^\top) = U^\top \Sigma_r U = A_{11}$. Since V^* contains the k_0 signal eigenvectors and $k - k_0$ noise eigenvectors, the eigenvalues of Σ_f^* are $\{\lambda_1, \dots, \lambda_{k_0}, \sigma^2, \dots, \sigma^2\}$ with σ^2 repeated $k - k_0$ times. The bound (27) follows directly from the definition of effective rank by comparing the normalized weights of the σ^2 terms to the total trace. This proves part (iii) and completes the proof of the theorem. $\square$

Theorem 6.1 explains why “illusory” complexity manifests as both (i) subspace instability (large Grassmann distances across resamples) and (ii) effective-rank collapse of fitted factor covariances. Flat or near flat curvature directions imply that small perturbations of Σ_r (e.g. sampling error) produce $O(1)$ rotations of the corresponding subspace components. Such components contribute little to $\text{erank}(\hat{\Sigma}_f)$ and produce large principal angle dispersion across resamples. In this sense, curvature degeneracy, subspace instability, and effective-rank collapse are three manifestations of the same structural phenomenon.

7 A Geometric Test of the Virtue of Complexity

We propose the following diagnostic framework:

- (i) Estimate models of increasing functional complexity.
- (ii) Compute effective rank of fitted factor covariance.
- (iii) Measure Grassmann distances across rolling windows.
- (iv) Evaluate out of sample pricing errors.

These diagnostics operationalize the geometric distinction between structural and illusory complexity. Complexity is virtuous if:

- Effective rank increases meaningfully,

- Subspace directions are stable,
- Out of sample performance improves.

Complexity is illusory if:

- Effective rank remains unchanged,
- Additional directions are unstable,
- Gains vanish out of sample.

8 Empirical Findings: Simulated Data

9 Empirical Findings: Historical Data

10 Conclusion

The debate over the “virtue of complexity” in asset pricing is fundamentally geometric.

Nonlinear exposure maps may expand parameter dimension without expanding intrinsic return dimension.

The economically relevant object is the return subspace. Grassmann geometry provides tools to measure intrinsic dimension, stability, and curvature.

Reframing the debate in geometric terms clarifies when complexity reflects structural discovery and when it reflects statistical illusion.

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